

BUSINESS REQUIREMENTS DOCUMENT

SQL-Driven Inventory Optimization System

Urban Retail Co.

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Document Title	Business Requirements Document — Inventory Optimization System
Project Name	SQL-Driven Inventory Optimization for Urban Retail Co.
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Status	Draft — Under Review
Stakeholders	Operations Manager, Procurement Team, Warehouse Manager, Finance Team

1. Executive Summary

Urban Retail Co. is a mid-sized retail business managing over 5,000 SKUs across multiple store locations. Despite steady revenue, the company has been operating on gut feel and reactive decisions when it comes to inventory products run out before anyone notices, slow-moving stock quietly eats up warehouse space and capital, and there is no single system to flag issues before they become expensive problems.

This document outlines the business requirements for an automated, SQL-driven inventory management and optimization system. The goal is simple: give the operations and procurement teams clear, real-time visibility into stock levels, demand patterns, and reorder needs — so they can stop reacting and start planning.

The system will track inventory across all stores, send automated low-stock alerts before stockouts happen, calculate data-backed reorder points, and surface KPIs through a Power BI dashboard. This is not just an analytics project — it is a foundation for smarter day-to-day business decisions.

2. Business Objectives

The core problems we are trying to solve are straightforward:

- Stockouts cost revenue.** Fast-moving products stock out, and sales walk out the door before anyone reorders.
- Overstock is quiet but expensive.** Slow-moving inventory locks up working capital and increases storage costs for weeks.

- **No visibility means reactive decisions.** No team has a clear picture of what is happening across stores and SKUs in real time.
- **Promotions are misfiring.** Promotions are being deployed during slow demand periods instead of proactively, which is reducing their effectiveness.

By the end of this project, the business should be able to:

- Receive automated alerts when any product approaches a critical stock threshold
- Know the right time and quantity to reorder — calculated from actual demand data, not guesswork
- See inventory performance across all 5 stores, 30 SKUs, and 5 categories in one dashboard
- Identify which products are carrying the most risk (fast movers with thin stock, dead stock sitting idle)
- Make better promotional decisions by understanding how demand actually responds

3. Project Scope

3.1 In Scope

- Inventory tracking across 5 store locations and 30 SKUs covering 5 product categories
- Automated low-stock alert system with configurable threshold logic
- Reorder point and reorder quantity calculation based on historical demand
- Demand pattern analysis — seasonal trends, product velocity, category performance
- Promotion impact analysis to understand what is and is not working
- KPI dashboard in Power BI with slicers for Store, Category, Region, and Promotions
- SQL-based data pipeline: cleaning, transformation, and KPI computation

3.2 Out of Scope

- Direct integration with supplier ordering systems or ERP platforms (Phase 2)
- Real-time POS or live inventory feed integration
- Customer-facing features or e-commerce inventory sync
- Predictive ML forecasting models (current MAPE of 15.9% will be addressed in a future iteration)

4. Stakeholders

Below are the key people this system is built for and the problem it solves for each of them:

Stakeholder	Role	Primary Need
Operations Manager	Project Sponsor	Single view of inventory health across all stores; early warning on stockout risks
Procurement Team	Primary User	Accurate reorder triggers and quantity recommendations to avoid manual guesswork

Warehouse Manager	Primary User	Visibility into slow-moving and excess stock to free up storage space
Finance Team	Secondary User	Working capital impact of overstock; holding cost reduction opportunities
Sales & Marketing	Secondary User	Understand how promotions are impacting actual demand patterns
IT / Data Team	Technical Owner	Maintain database schema, run SQL pipelines, and manage Power BI refresh

5. Current State (As-Is)

Right now, Urban Retail Co. has no proactive system for managing inventory decisions. Here is what the data tells us about the current situation:

1.23% Stockout Rate (Inventory < 50 units)	2.51% Overstock Rate (Inventory > 250 units)
15.9% Demand Forecast MAPE (Error Rate)	0.67 Avg Inventory Turnover (Low)
-16.2% Promo Sales Impact (Underperforming)	$r \approx 0.99$ Pricing vs Competitors (Purely Reactive)

What this means in plain terms:

- Electronics holds the most inventory but has the lowest turnover (0.66) — clear overstocking in that category
- Top-performing products (P0057, P0125) sell ~33% more than bottom performers, yet reorder logic treats them the same
- Winter demand is 15% higher than Spring, but ordering patterns do not reflect this seasonal shift
- Promotions are being run during low-demand periods — they show a negative impact of -16.2% on sales vs. non-promo periods
- Discounts show no measurable impact on volume — the current discount strategy needs a rethink
- Regional performance is relatively balanced (< 3% spread), so the problems are operational, not geographic

The root cause: decisions are made after the damage is already done. There is no system catching these patterns early.

6. Future State (To-Be)

After implementing this system, here is what the day-to-day should look like:

- The procurement team gets automated alerts when any SKU crosses its low-stock threshold — before it hits zero
- Reorder quantities are calculated from actual demand data and safety stock formulas, not from someone's memory
- The operations manager opens the Power BI dashboard every Monday and immediately sees which stores need attention
- Finance can see exactly how much capital is tied up in slow-moving stock and where it can be released
- Marketing can see demand response curves before deciding when and where to run the next promotion

The shift we are enabling: from reactive firefighting to proactive inventory management — backed by data that is always current.

7. Functional Requirements

These are the specific capabilities the system must deliver:

ID	Requirement	Priority
FR-01	System shall track real-time stock levels across all stores and SKUs	High
FR-02	System shall flag any SKU where inventory falls below 50 units as a low-stock alert	High
FR-03	System shall calculate reorder point using lead time and average daily demand	High
FR-04	System shall compute reorder quantity based on EOQ or safety stock logic	High
FR-05	System shall segment products into Fast / Moderate / Slow movers based on turnover	High
FR-06	System shall compute inventory turnover rate per SKU, category, and store	Medium
FR-07	System shall track and report demand forecast accuracy (MAPE)	Medium
FR-08	System shall measure and report the sales impact of promotional campaigns	Medium
FR-09	Power BI dashboard shall display KPIs with slicers for Store, Category, Region, and Promotions	High
FR-10	System shall produce a weekly executive summary report for the Operations Manager	Low

8. Non-Functional Requirements

Category	Requirement
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Performance	SQL queries must return results within 5 seconds for standard KPI reports on full dataset
Scalability	System must support expansion to 10+ stores and 10,000+ SKUs without redesign
Data Freshness	Inventory data must be updated at minimum once per day; alerts must trigger within the same day
Reliability	Dashboard uptime should be at least 99% during business hours (9 AM – 7 PM)
Usability	Non-technical users (procurement, ops) must be able to navigate the dashboard without training
Data Integrity	No negative inventory values; all SKU IDs and store IDs must be validated before processing
Security	Access to raw data restricted to IT and Data teams; dashboard access controlled by role
Maintainability	SQL scripts must be modular and documented so any team member can update thresholds

9. Assumptions & Constraints

9.1 Assumptions

- Historical sales data from January – December 2022 (16,383 records) is accurate and representative of typical demand patterns
- The 5-store, 30-SKU, 5-category dataset is sufficient for building and validating the reorder logic
- Stakeholders will provide feedback on dashboard designs within one week of each review
- The IT team will manage data refresh schedules once the pipeline is handed over
- Weather data will not be used as a demand predictor (analysis confirmed no significant correlation)

9.2 Constraints

- No budget allocated for ERP integration or real-time POS feeds in this phase
- All analysis is based on historical data — the system does not support live inventory feeds yet
- Power BI dashboard is hosted on internal SharePoint; external access requires IT approval
- Forecast accuracy (MAPE 15.9%) is a known limitation; ML-based forecasting is out of scope for v1

10. Risks & Dependencies

Risk / Dependency	Likelihood	Mitigation
Data quality issues in raw inventory feed (negative values, missing IDs)	Medium	Validation rules built into SQL cleaning layer; anomalies flagged for manual review

Stakeholder availability for UAT and dashboard sign-off	Medium	Schedule review sessions in advance; async feedback via shared document
Forecast MAPE remains high (>15%) even after reorder logic is applied	High	Acknowledge as known v1 limitation; plan ML forecasting module for Phase 2
Power BI refresh failures causing stale dashboard data	Low	Set up email alerts on refresh failures; IT team on escalation path
Scope creep — requests for ERP or live POS integration mid-project	Medium	Document and park these requests formally in a Phase 2 backlog

11. Acceptance Criteria

The system will be considered complete and ready for handover when all of the following conditions are met:

- All 6 SQL modules (stock availability, low-stock alerts, inventory turnover, demand forecasting, reorder point, reorder quantity) run without errors on the full dataset
- Low-stock alerts correctly identify all SKUs below the 50-unit threshold in test scenarios
- Reorder point and quantity calculations match manually verified results for at least 5 test SKUs
- Power BI dashboard loads in under 5 seconds and all slicers (Store, Category, Region, Promotions) filter correctly
- Operations Manager and Procurement Lead sign off on dashboard usability after a 30-minute walkthrough
- Executive summary report is reviewed and approved by the Operations Manager
- All SQL scripts are documented with inline comments and handed over to the IT team

12. Sign-Off & Approvals

This document requires review and approval from the following stakeholders before development begins:

Name	Role	Sign-Off Date
[Operations Manager]	Project Sponsor	_____
[Procurement Lead]	Primary User	_____
[IT / Data Lead]	Technical Owner	_____
Md Sahil Ansari	Business Analyst	April 2025